

# Seminar 13 - Algebră

## Transformări liniare și matrice

- Fie  $U, V$   $K$ -sp. vectoriale,  $K$  corp com.  $\dim U = m, \dim V = n$
- Fie  $u = (u_1, \dots, u_m)$  bază pt  $U$ ,  $v = (v_1, \dots, v_n)$  bază pt  $V$
- Fie  $f: U \rightarrow V$   $K$ -liniară,  $[f]_{u,v} \in M_{n,m}(K)$

$$[f]_{u,v} = \begin{bmatrix} | & | & \dots & | \\ f(u_1) & f(u_2) & \dots & f(u_m) \\ | & | & \dots & | \end{bmatrix}$$

Coord. lui  $f(u_1)$  în bază  $v$       Coord. lui  $f(u_2)$  în bază  $v$       Coord. lui  $f(u_m)$  în bază  $v$

- Fie  $u' = (u'_1, \dots, u'_m)$  bază pt  $U$   
 $v' = (v'_1, \dots, v'_n)$  bază pt  $V$

$$[f]_{u',v'} = ?$$

$$u' = uS, \quad v' = vT$$

$S, T$  matrice

$$\begin{array}{ccc} U: & u & \xrightarrow{S} & u' & (u' = uS) \\ f \downarrow & [f]_{u,v} & & [f]_{u',v'} \\ V: & v & \xrightarrow{T} & v' & (v' = vT) \end{array}$$

$\curvearrowright \quad T^{-1}$

$$\Rightarrow [f]_{u',v'} = T^{-1} \cdot [f]_{u,v} \cdot S$$

- formula de schimbare a bazei pt mat. transformării

ex: a) Fie  $f \in \text{Aut } \mathbb{R}^2$  și  $h: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  liniară

$$h(x, y) = (x \cos \varphi - y \sin \varphi, x \sin \varphi + y \cos \varphi)$$

este automorfism. Scrieți mat. lui  $h$  în bază canonică a lui  $\mathbb{R}^2$



$$e) f: \mathbb{R}^2 \rightarrow \mathbb{R}^2, f(x,y) = (x, -y), [f]_e, [g]_e, [f \circ g]_e, [f+g]_e \\ g: \mathbb{R}^2 \rightarrow \mathbb{R}^2, g(x,y) = (-x, y) [g \circ f]_e$$

Sol: a) h este o transformare liniară

$$h(e_1) = h(1,0) = (\cos t, \sin t) \\ h(e_2) = h(0,1) = (-\sin t, \cos t) \Rightarrow [h]_e = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$$

$$h \text{ automorfism} \in \text{Aut}(\mathbb{R}^2) \Leftrightarrow [h]_e \in GL_2(\mathbb{R}) \Leftrightarrow \begin{vmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{vmatrix} = 1$$

$$e) f(e_1) = f(1,0) = (1,0) \\ f(e_2) = f(0,1) = (0,-1) \Rightarrow [f]_e = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}; \det [f]_e \neq 0$$

$$g(e_1) = g(1,0) = (-1,0) \\ g(e_2) = g(0,1) = (0,1) \Rightarrow [g]_e = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}; \det [g]_e \neq 0$$

$$[f \circ g]_e = [f]_e \cdot [g]_e = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} \quad f, g \in \text{Aut}(\mathbb{R}^2)$$

$$[f+2g]_e = [f]_e + 2[g]_e$$

$$[g \circ f]_e = [g]_e \cdot [f]_e$$

ex: Fie  $f: \mathbb{R}^2 \rightarrow \mathbb{R}^3, f(x,y) = (x+y, 2x-y, 3x+2y)$  - Ar  
 $f$  transformare lin,  $V = ((1,2), (-2,1))$  bază pt  $\mathbb{R}^2$   
 $V' = ((1,-1,0), (-1,0,1), (1,1,1))$  bază pt  $\mathbb{R}^3$   
 $[f]_{V,V'} = ?$

Sol:  $\begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} = 5 \neq 0 \Rightarrow V$  bază pt  $\mathbb{R}^2$ ;  $\begin{vmatrix} 1 & -1 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} = -3$   
 $\Rightarrow V$  bază pt  $\mathbb{R}^3$



$$g(v_1) = \alpha_{11} v_1' + \alpha_{12} v_2' + \alpha_{13} v_3' \rightarrow \text{comb lin.}$$

$$g(1,2) = \alpha_{11}(1,1,0) + \alpha_{12}(-1,0,1) + \alpha_{13}(1,1,1)$$

$$(3,0,7) = (\alpha_{11} - \alpha_{12} + \alpha_{13}, -\alpha_{11} + \alpha_{13}, \alpha_{12} + \alpha_{13})$$

$$\Rightarrow \begin{cases} \alpha_{11} - \alpha_{12} + \alpha_{13} = 3 \\ -\alpha_{11} + \alpha_{13} = 0 \\ \alpha_{12} + \alpha_{13} = 7 \end{cases} \Rightarrow \alpha_{11} = \frac{10}{3}$$

$$3\alpha_{13} = 10 \Rightarrow \alpha_{13} = \frac{10}{3}$$

$$[g]_{v,v'} = \begin{pmatrix} 10/3 & 5/3 \\ 11/3 & -2/3 \\ 10/3 & -10/3 \end{pmatrix}$$

$$g(v_2) = \alpha_{21} v_1' + \alpha_{22} v_2' + \alpha_{23} v_3'$$

$$g(2,1) = \alpha_{21}(1,1,0) + \alpha_{22}(-1,0,1) + \alpha_{23}(1,1,1)$$

$$\left. \begin{aligned} g(2,1) &= \alpha_{21}(1,1,0) + \alpha_{22}(-1,0,1) + \alpha_{23}(1,1,1) \\ (-1, -5, 4) &= \end{aligned} \right\} \Rightarrow$$

$$\Rightarrow \begin{cases} -1 = \alpha_{21} - \alpha_{22} + \alpha_{23} \\ -5 = -\alpha_{21} + \alpha_{23} \\ -4 = \alpha_{22} + \alpha_{23} \end{cases}$$

$$-10 = 3\alpha_{23} \Rightarrow \alpha_{23} = -\frac{10}{3}$$

$$\alpha_{21} = \frac{5}{3}$$

$$\alpha_{22} = -\frac{2}{3}$$

II  $e = (e_1, e_2)$  base canonica in  $\mathbb{R}^2$

$e' = (e'_1, e'_2, e'_3)$  k - in  $\mathbb{R}^3$

$$\Rightarrow g(e_1) = g(1,0) = (1,2,3)$$

$$g(e_2) = g(0,1) = (1,-1,2)$$

$$\Rightarrow [g]_{e,e'} = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ 3 & 2 \end{bmatrix}, \quad \begin{aligned} v &= eS \\ v' &= e'T \end{aligned}$$



unde  $S = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$  ead vectorial (placem de la baza  
comune)

$$T = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \Rightarrow T^{\#} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \Rightarrow T^* = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & 2 \\ -2 & -2 & 2 \end{bmatrix}$$

$$\{f\}_{v,v'} = T^{-1} \cdot \{f\}_{e,e'} \cdot S$$

ex: Fie  $V = (v_1, v_2, v_3, v_4)$  baza e lui  $\mathbb{R}^4$

$$u_1 = v_1, u_2 = v_1 + v_2, u_3 = v_1 + v_2 + v_3, u_4 = v_1 + v_2 + v_3 + v_4$$

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 3 & 0 & -1 & 2 \\ 2 & 5 & 3 & 1 \\ 1 & 2 & 1 & 3 \end{bmatrix}, f \in \text{End}(\mathbb{R}^4)$$

a)  $u_1, u_2, u_3, u_4$  baza pt  $\mathbb{R}^4$

b) Stind c  $\{f\}_v = A, \{f\}_u = ?$

c) Stind c  $\{f\}_u = A, \{f\}_v = ?$

d)  $u = vS, S = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}; \det S = 1 \neq 0 \Rightarrow u \text{ baza pt } \mathbb{R}^4$   
(S: matricea de trecere de la  $v$  la  $u$ )

e)  $\mathbb{R}^4$   
 $f \downarrow$   
 $\mathbb{R}^4$

$$\begin{array}{ccc} v & \xrightarrow{S} & u \\ \{f\}_v & & \{f\}_u \\ v & \xrightarrow{S} & u \\ \underbrace{\quad \quad \quad}_{S^{-1}} & & \end{array}$$

$$\boxed{\{f\}_u = S^{-1} \cdot A \cdot S = \dots}$$

f)  $\boxed{\{f\}_v = S \cdot A \cdot S^{-1} = \dots}$



ex: Für  $v, v' \in \mathbb{R}$ -Vekt,  $(a_1, a_2, a_3) = a$  Basis für  $V$   
 $(b_1, b_2, b_3) = b$  Basis für  $V'$   
 Für  $f: V \rightarrow V'$   $\mathbb{R}$ -linear

$$[f]_{a,b} = \begin{bmatrix} -1 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

a)  $f(v) = ?$ ,  $v \in V$

b)  $\dim \text{Im} f$ ,  $\dim \text{Ker} f = ?$

c)  $a' = (a_1, a_1 + a_2, a_1 + a_2 + a_3)$   
 $b' = (b_1, b_1 + b_2, b_1 + b_2 + b_3)$  ,  $[f]_{a',b'} = ?$

Sol:  $v \in V \Rightarrow v = x_1 a_1 + x_2 a_2 + x_3 a_3$ ,  $x_1, x_2, x_3 \in \mathbb{R}$

$$f(v) = f(x_1 a_1 + x_2 a_2 + x_3 a_3) = x_1 f(a_1) + x_2 f(a_2) + x_3 f(a_3)$$

$$f(a_1) = -b_1 + b_2 + 0 \cdot b_3 = x_1(-b_1 + b_2) + x_2 \cdot 0 + x_3(b_1 + b_2)$$

$$f(a_2) = 0$$

$$f(a_3) = b_1 - b_2$$

$$= b_1(-x_1 + x_3) + b_2(x_1 - x_3)$$

$\dim = \text{rang}$  einer Matrix

$\hookrightarrow \text{transf. lin.}$

b)  $\dim \text{Im} f = \text{rang } [f]_{a,b} = 2$

$$\dim \text{Ker} f + \underbrace{\dim \text{Im} f}_2 = \underbrace{\dim V}_3 \Rightarrow \dim \text{Ker} f = 1$$

c)  $a' = a \cdot S$ ,  $S = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

$$b' = b \cdot T$$
,  $T = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

$$[f]_{a',b'} = T^{-1} \cdot [f]_{a,b} \cdot S = S^{-1} [f]_{a,b} S = \dots$$



ex: Sei  $V, V', \mathbb{R}$ -Vekt.,  $(v_1, v_2, v_3) = v$  Basis in  $V$   
 $(v'_1, v'_2, v'_3) = v'$  Basis in  $V'$

$f: V \rightarrow V', \mathbb{R}$ -linear

$$[f]_{v, v'} = \begin{bmatrix} 0 & -1 & 5 \\ 1 & 0 & 0 \\ 0 & 1 & -5 \end{bmatrix}$$

a) Dimensionen & Basis & Basis pt. Img, Kerf

b)  $[f]_{v, v'} = ?$  da  $V = \mathbb{R}^3$   $v = e$  Basis annehmen

$$v'_1 = (1, 0, 0), v'_2 = (0, 1, 0), v'_3 = (0, 0, 1)$$

c)  $f(x) = ?$ ,  $x = 2v_1 - v_2 + 3v_3$  im Koordinatenfeld

a)  $\dim \text{Img} = \text{rang } [f]_{v, v'} = 2$

$$f(v_1) = v'_2$$

$$f(v_2) = -v'_1 + v'_3 \Rightarrow (v'_2, -v'_1 + v'_3) \text{ Basis pt. Img}$$

$$\Rightarrow \dim \text{Kerf} = \dim V - \dim \text{Img} = 3 - 2 = 1$$

$$\text{Oder } c_3 = -5c_2 \Leftrightarrow f(v_3) = -5f(v_2) \Leftrightarrow f(v_3) + 5f(v_2) = 0 \Leftrightarrow$$

$$\Leftrightarrow f(v_3 + 5v_2) = 0 \Rightarrow v_3 + 5v_2 \in \text{Kerf}$$

$\Rightarrow \{v_3 + 5v_2\}$  Basis pt. Kerf

$$\text{Setze } \forall x \in V \Rightarrow x = x_1 v_1 + x_2 v_2 + x_3 v_3$$

$$x \in \text{Kerf} \Leftrightarrow f(x_1 v_1 + x_2 v_2 + x_3 v_3) = 0 \Leftrightarrow$$

$$\Leftrightarrow \underbrace{x_1}_{=0} f(v_1) + \underbrace{x_2}_{=5x_3} f(v_2) + \underbrace{x_3}_{\in \mathbb{R}} f(v_3) = 0$$

da  $x$  im Basis  $v$

$$f(x) = 0 \Leftrightarrow [f]_{v, v'} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \dots$$

$$\begin{cases} x_1 = 0 \\ x_2 = 5x_3 \\ x_3 \in \mathbb{R} \end{cases}$$

$$\text{Sol: Kerf} = \{5x_3 v_2 + x_3 v_3\} = \langle 5v_2 + v_3 \rangle$$



$$b) v' = e' \cdot T ; T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\begin{array}{ccccc} \mathbb{R}^3 = V & : & V = e & \xrightarrow{J_3} & V = e \\ \downarrow f & & \downarrow [f]_{V,V'} & & \downarrow [f]_{V,e'} \\ V' & & V' & \xrightarrow{T^{-1}} & e' \end{array}$$

$$[f]_{V,e'} = T \cdot [f]_{V,V'} \cdot T^{-1} = T [f]_{V,V'}$$

$$c) f(x) = [f]_{V,e'} \cdot \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 & 4 & 5 \\ 1 & 0 & 0 \\ 0 & -4 & -5 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 11 \\ 2 \\ -11 \end{bmatrix}$$